Mateo Jimenez-Sotelo

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Education

National Autonomous University of Mexico (UNAM) Cuernavaca, Mexico
B.Sc. in Genomic Sciences, GPA: 9.54/10 Aug 2024 - present
Relevant Coursework: Statistics, Calculus, Linear Algebra, Differential Equations, Mathematical Modelling, Data Science, Programming, Bioinformatics, Genetics, Human & Population Genomics, Transcriptomics, Epigenetics, Molecular & Cell Biology

Research experience

Center for Genomics Sciences, UNAM Cuernavaca, Mexico
Undergraduate researcher Dec 2024 - Present
PI: Julio A. Freyre-Gonzalez, Systems Biology Lab.

- **Causal Discovery in Gene Regulatory Networks**
  - Investigating causal discovery methods for reconstructing gene regulatory networks from observational transcriptomic data.
  - Evaluating graph-based and machine learning approaches, including additive noise models, SCORE, diffusion models (DiffAN), and transformer-based architectures.
  - Integrating regulatory network resources such as *Abasy Atlas*, *i-Modulons* and the *Natural Decomposition Approach* for benchmarking and biological interpretation.
- **NLP for transcriptomic metadata harmonization**
  - Developing transformer-based NLP pipelines to standardize RNA-seq and microarray metadata from Gene Expression Omnibus (GEO).
  - Evaluating large language models for biological metadata normalization using controlled vocabularies and ontology-aware representations.
  - Implementing preprocessing, embedding generation, and semantic similarity analyses using PyTorch, Hugging Face Transformers, and pandas.

International Laboratory for Human Genome Research, UNAM Juriquilla, Mexico
Undergraduate researcher Jan 2025 - May 2026
PI: Maria Avila-Arcos, Population & Evolutionary Genomics Lab.

- Performed principal component analysis (PCA) and ADMIXTURE analyses of ancient and present-day genomic datasets.
- Integrated ancient individuals from *La Conchita* and *Hospital San José de los Naturales* with reference panels from the 1000 Genomes Project and the Mexican Biobank.
- Processed genotype and sequencing datasets using PLINK, SAMtools, BCFtools, EIGENSOFT (smartpca), ADMIXTOOLS, and Pong.
- Generated visualizations and contributed to the interpretation of ancestry patterns in colonial-era individuals from New Spain.
- Co-authored a manuscript on pathogen introductions to Mexico following the European Conquest; manuscript in preparation for submission (planned submission: November 2026).

Skills

Programming: Python, Bash, R, Julia | **Working knowledge**: C/C++, SQL
Bioinformatics: Population genomics, Ancient DNA, RNA-seq analysis, Differential expression, Functional enrichment, Gene regulatory networks, Structural bioinformatics, Genome assembly
Scientific Computing: Linux, Git, Nextflow, Conda, HPC environments, Quarto
Libraries: PyTorch, Transformers, Pandas, NumPy, Bioconductor, Scikit-learn
Genomics tools: PLINK, EIGENSOFT, ADMIXTOOLS, ADMIXTURE, SAMtools, BCFtools, Pong
Languages: Spanish (native), English (C1), German (beginner)

Teaching experience

National Autonomous University of Mexico - Genomic and Biomedical Sciences programs Cuernavaca, Mexico
***Teaching assistant* - Principles of Programming** Aug 2026 - Dec 2026

- Assisting with course instruction, student mentoring, assessment, grading, and weekly programming tutorials.

Selected projects

- Nextflow RNA-seq preprocessing workflow
- Transcriptomic analysis of brain aging in *Macaca mulatta*
- Comparative genomics of *Babesia divergens*
- ChIP-seq peak-to-FASTA software for transcription factor binding-site extraction
- Wright-Fisher Genetic Drift Simulation

Additional training

Biology Olympiad Training Program - UNAM Mexico City, Mexico
Academic Trainee May 2023 - August 2023
Moises O. Fiesco-Roa, Embryology and Genetics Department, Faculty of Medicine, UNAM.

- Completed interdisciplinary coursework across multiple UNAM institutions, including Faculties of Medicine and Sciences and Instituto Nacional de Pediatría (INP).
- Training topics included general biology, genetics, neurophysiology, plant biology, population genetics, mycology, ornithology, developmental biology, and cellular signaling.

Awards

- Graduated with Honors, Colegio Madrid (CCH curriculum), 2024
  Final GPA: 9.70/10
- Gold Medalist, Mexico City Biology Olympiad (OCaBi), 2023

Professional development

- **Large Language Models and HF ecosystem** - Hugging Face, Feb 2026
- **Fundamentals of Deep Learning** - Deep Learning Institute (NVIDIA) and the Institute of Biotechnology (UNAM), Aug 2025.